

PROG8435 – Data Analysis, Modeling and Algorithms

Assignment 5

Classification

DUE BEFORE 11:59 pm, APRIL 14,

1. Submission Guidelines

All assignments must be submitted via the econestoga course website before the due date into the assignment folder.

You may make multiple submissions, but only the most current submission will be graded.

SUBMISSIONS

In the Assignment 4 Folder submit:

1. Your *.R file. This file must have all your comments and answers to the questions. If you do not include your output, I will *not* be running your code to generate it. I may, however run the code to verify the results.
2. The *.pdf or *.doc file that is produced from your code.

DO NOT PUT THE DOCUMENTS IN TO A ZIP FILE!

PLEASE NOTE: The marks on the assignment are generally awarded 50% for the actual R code and calculations and 50% for interpretation and demonstration that you understand what you have done.

EXAMPLES: The example output provided is simply to demonstrate what a typical submission might look like. You can use it as a basis, but your submission must be in your own words. Submissions that simply “cut and paste” my example commentary will be marked 0.

All variables in your code must abide by the naming convention [variable_name]_[initials]. For example, my variable for State would be State_DM.

THIS IS AN INDIVIDUAL ASSIGNMENT. UNAUTHORIZED COLLABORATION IS AN ACADEMIC OFFENSE AS IS DIRECT ‘CUTTING AND PASTING’ FROM OTHER SOURCES. Please see the Conestoga College Academic Integrity Policy for details.

Remember the discussion forums on eConestoga are a great place to ask questions.

2. Grading

This assignment will be marked out of 30 and is worth 12.5% of your total grade in the course.

Assignments submitted after 11:59 pm will be reduced 20%. Assignments received after 8:00am the morning after the due date will receive a mark of 0%.

Assignments which do not follow the submission instructions may have marks deducted.

3. Data

Each student will be using the study dataset:

STUDY DATASET:

PROG8435-Assign04-24W.txt

Appendix one contains a data dictionary for the study file.

4. Background

This is the same background and study file as the one used in Assignment 4.

You will be working with a dataset which contains a count of public bicycles rented per hour in the Seoul Bike Sharing System, with corresponding weather data and holiday information.

“Currently Rental bikes are introduced in many urban cities for the enhancement of mobility comfort. It is important to make the rental bike available and accessible to the public at the right time as it lessens the waiting time. Eventually, providing the city with a stable supply of rental bikes becomes a major concern. “

Your task will be to use a variety of classifiers to determine the factors which contribute to, and help predict, ‘High Use’ of bicycles (High Use is defined as usage above 1700). As always, imagine this analysis will be generating a report that will be presented to a client.

All of the tasks have been demonstrated in class. A careful review of your notes from the lectures should give you everything you need to complete these tasks.

5. Assignment Tasks

NOTE – For each step/task, include sufficient commentary in your submission to demonstrate understanding of the steps you have taken.

Nbr	Description	Marks
1	Preliminary Data Preparation 1. Rename all variables with your initials appended (just as was done in previous assignments). Remember that any variables you subsequently create need to have your initials appended. 2. Delete the observation with Temp > 155 using the following code: <code>ClssAssign <- ClssAssign[!ClssAssign\$Temp > 150,]</code> NOTE – Your variable names will be different of course! Also, make sure you separate it into train and test sets 3. Create a new variable in the dataset called HB_[Initials] which will have a value of 1 if RBC ≥ 1700 and 0 otherwise. If you have forgotten how to do this, the code to accomplish it is included in the appendix. Remember to also delete the RBC variable after this.	3
2	Exploratory Analysis 1. Correlations: Create correlations (as demonstrated) and comment on what you see. Are there co-linear variables?	1
3	Model Development As demonstrated in class, create two logistic regression models. 1. A full model using all of the variables. 2. An additional model using backward, stepwise or manual selection. Your goal is to create the best model possible. For each model, interpret and comment on the main measures we discussed in class: (1) Fisher iterations (2) AIC (3) Residual Deviance (4) z-values (5) Parameter Co-Efficients 3. As demonstrated in class, analyze the output for any significantly influential datapoints. Are there any? 4. Based on your preceding analysis, recommend which model should be selected and explain why.	1 1 3 1 1
PART B		
In this section, all classifiers should be built using HB_ [Initials] as the dependant variable and the remaining variables as the independent variables.		
1	Logistic Regression – stepwise 1. As above, use the step option in the glm function to fit the model (using stepwise selection).	1 2

	- Summarize the results in Confusion Matrices for both train and test sets. - Calculate the time (in seconds) it took to fit the model and include this in your summary.	1
2	Naïve-Bayes Classification - Use all the variables in the dataset to fit a Naïve-Bayesian classification model. - Summarize the results in Confusion Matrices for both train and test sets. - Calculate the time (in seconds) it took to fit the model and include this in your summary.	1 2 1
3	Recursive Partitioning Analysis - Use all the variables in the dataset to fit a recursive partitioning classification model. - Summarize the results in Confusion Matrices for both train and test sets. - Calculate the time (in seconds) it took to fit the model and include this in your summary.	1 2 1
4	Compare All Classifiers For all questions below please provide evidence. - Which classifier is most accurate? - Which classifier seems most consistent (think train and test)? - Which classifier is most suitable when processing speed is most important? - Which classifier minimizes false positives? - In your opinion, classifier is best overall?	5
5	Professionalism and Clarity	2

APPENDIX ONE: STUDY FILE DATA DICTIONARY

Variable	Description
RBC	Rented Bike Count: Number of bicycles rented in that time period.
HR	Hour of the Day (0 to 23) (0: 12:00am to 12:59am, 1: 1:00am to 1:59am, etc.)
SR	Solar Radiation (M_j/m^2)
WS	Wind Speed (m/s)
SF	Snow Fall (cm)
Hol	Whether the day was a National Holiday
Humid	Humidity (%)
RF	Rain Fall (mm)
Temp	Temperature (C)
Seas	Season of the Year

Example Code for outcome

```
df$HB_DM <- as.factor(ifelse(df$RBC_DM >= 1700, 1,0))
```